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The ion channels: one function, twenty-five origins

A review of the ion-channel superfamilies, their folds, their evolution and what the sequence databases do and do not record about them.

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Summary

An ion channel is a gated aqueous pore. That is a description of what a protein does, not of where it came from, and the distinction organises everything that follows. The proteins the term covers do not form a family, a superfamily, or any other clade: they are on the order of twenty-five independent inventions, built on at least ten unrelated folds, that arrived at the same solution to the same physical problem. There is no alignment containing a nicotinic acetylcholine receptor and a potassium channel, and therefore no tree.

This review is organised around that fact. We take the folds one at a time — the P-loop superfamily that supplies nearly half the human channel genes, the four unrelated ligand-gated superfamilies, the anion channels, the large pores, the mechanosensors and the channels of intracellular membranes — and ask of each what is genuinely shared and what merely looks shared. We then turn to the two questions a modern census has to answer before it can count anything: what the annotation databases actually record about these proteins, and where their coverage fails.

The second question turns out to be sharper than expected. Domain models were built family by family from the best-studied members, and they behave accordingly: the canonical pore model PF00520 is carried by twenty catalogued families including a phosphatase that is not a channel, while most of the TRP families carry no copy of it at all. Three of the four prokaryotic outgroups that root the major superfamily trees cannot be found by the domain rules that define their own superfamilies. These are not database errors. They are what a set of models built from well-studied proteins looks like when it is asked about the proteins nobody studied.

Key points

- **“Ion channel” is a functional class, not a clade.** Any statement of the form “ion channels evolved...” is either about one superfamily or is wrong.
- **The human channelome is lopsided.** Of ~320 pore-forming genes, 143 are in the P-loop superfamily and 79 of those are potassium channels; the remaining 177 are spread across twenty-four superfamilies.
- **Convergence is the rule.** Potassium selectivity evolved at least twice; vertebrates run two unrelated gap-junction systems; mechanosensitivity has been invented at least six times.
- **Selectivity and gating are not recoverable from sequence family.** Cys-loop charge selectivity flips with three substitutions; AMPA-receptor calcium permeability is set by RNA editing.
- **Domain annotation does not partition the subject the way the textbooks do,** and a census built on the obvious accessions silently loses whole divisions.

1. What is being counted

The modern study of ion channels begins with a quantitative model that required no molecules at all. Hodgkin and Huxley described the squid axon action potential as the sum of independent, voltage-dependent sodium and potassium conductances, and the description was so complete that the proteins responsible could be inferred long before any was isolated [1]. Single-channel recording made the discrete conducting unit visible directly [2], and molecular cloning then produced the sequences: the electric-ray acetylcholine receptor [3], the eel sodium channel [4], the *Drosophila* Shaker potassium channel [5,6]. Within a decade the field had a structure — KcsA, at 3.2 Å, showing exactly how a filter of backbone carbonyls discriminates potassium from sodium [7].

That history explains a persistent difficulty. Each family was cloned, named and modelled on its own, and the resulting nomenclature and annotation reflect the order of discovery rather than any underlying structure. The term “ion channel” was never a phylogenetic claim; it was a description of behaviour that happened to be shared.

1.1 A definition, and its six edges

Throughout this review an ion channel is:

a protein assembly forming a **gated aqueous pore** that conducts ions across a membrane **down their electrochemical gradient**, without a coupled conformational transport cycle.

Every word does work, and each marks a boundary that published counts of the human channelome — which range from roughly 240 to roughly 400 — draw differently:

1. **Auxiliary subunits.** KCNE1 is 129 residues with one transmembrane helix and no pore, and it converts KCNQ1 into the cardiac IKs channel. The Cav β , $\alpha 2\delta$ and γ subunits, the Nav β subunits, the sulfonylurea receptors, STIM1 and barttin are likewise essential and likewise not pores. Counting them inflates the human total by roughly 15 %.
2. **Transporters on a channel fold.** Half the CLC family are 2Cl⁻/H⁺ antiporters, including the bacterial prototype [8]. They are CLCs by descent and not channels by mechanism.
3. **Scramblases on a channel fold.** TMEM16A and TMEM16B are chloride channels; most of their relatives are phospholipid scramblases that also pass ions [9].
4. **Large pores.** Connexins, pannexins, LRRC8 and CALHM conduct ions through a gated aqueous pathway wide enough to pass ATP. That the pore is wide does not make it not a channel.
5. **Non-ion channels.** Aquaporins are gated, selective, ancient and exhaustively characterised, and they conduct water. They mark the outer edge of the definition and are excluded by it.
6. **Viral and prokaryotic channels.** Influenza M2, the bacterial mechanosensitive channels and KcsA are channels by every criterion, and they are the structural prototypes on which the rest of the field is built.

None of these six answers is obviously right. What matters is that a count without them stated is not interpretable, and that the things excluded should be excluded by a recorded decision rather than by never having been considered.

1.2 What a census has to survive

Two further problems are specific to counting this subject rather than describing it.

The first is that **gene symbols group pores with their accessory subunits**. *KCNE**, *KCNMB**, *CACNB**, *CACNG**, *SCN*B* and the 25 *KCTD** genes all sort inside the channel symbol space, and none is a pore; the *CACNG* symbols are worse than uninformative, since *CACNG2–8* are AMPA-receptor auxiliary subunits with a claudin fold and no role in calcium channels. A census that reads names counts the wrong things, and an unnamed gene model in a newly assembled genome — the case a census exists for — has no name to read.

The second is that **domain annotation is not a partition of the subject**. Section 12 takes this up in detail with measurements; the short version is that the canonical pore model is carried by families that are not channels, and is absent from families that are.

2. The folds

A useful way to organise the subject is by the architecture of the pore itself, because that is what is genuinely inherited. At least ten unrelated folds solve the problem, and the boundaries between them are the boundaries beyond which no alignment is meaningful.

The P-loop (voltage-gated-like) fold. Four subunits, or four repeats of one chain, each contributing two transmembrane helices and a re-entrant loop that dips into the membrane from the outside. The loop supplies the selectivity filter; the last helix lines the inner cavity and carries the gate. This is the fold of the potassium, sodium and calcium channels, the TRP families, the cyclic-nucleotide-gated and hyperpolarisation-activated channels, the intracellular calcium-release channels and the two-pore channels [7,10]. It is also the only channel fold in this review large enough to need a within-superfamily phylogeny of its own.

The pentameric Cys-loop fold. Five subunits around a central axis, each with a β -sandwich extracellular ligand-binding domain over a four-helix transmembrane bundle; the second helix of each subunit lines the pore. Known from the *Torpedo* receptor [11], from the soluble acetylcholine binding protein that turned out to be a complete ligand-binding domain without a channel [12], and from bacterial members that lack the disulfide the superfamily is named after [13,14].

The iGluR fold. Four subunits, a clamshell ligand-binding domain derived from a bacterial periplasmic binding protein, and — pointing the other way — a pore module that is an inverted P-loop [15]. The relationship to potassium channels is structural and is confirmed by a cyanobacterial glutamate receptor with an intact potassium filter [16].

The trimeric folds. P2X receptors are trimers with a large disulfide-rich ectodomain and both termini inside [17]; the DEG/ENaC channels are trimers of a different fold entirely [18,19].

The CLC fold. An antiparallel homodimer with two entirely independent pores, one per subunit — inferred from noise analysis three decades before the structure confirmed it [20].

The TMEM16 fold. Ten transmembrane helices per subunit with a membrane-facing groove between TM4 and TM6; open to the lipid in the scramblases, closed into a pore in the channels [21]. The OSCA mechanosensitive channels and, on present evidence, the TMC proteins share this fold without sharing detectable sequence [22].

The remaining folds are each a family of their own: the three-bladed Piezo propeller with 38 transmembrane helices per subunit [23]; the 19-stranded β -barrel of VDAC [24]; the hexameric connexin hemichannel [25] and the unrelated innexin/pannexin/LRRC8 architecture [26,27]; the CALHM large pore [28]; MscL and MscS, which are unrelated to each other [29,30]; the ABC fold of CFTR [31]; the double-barrelled otopetrin fold [32]; the voltage-sensor-only architecture of Hv1 [33,34]; and TMEM175, whose potassium selectivity is achieved without a filter at all [35].

2.1 What “unrelated” means here

Three of these relationships deserve care, because the word covers three different situations.

Undetectable but probably real. TMEM16, OSCA and TMC share a fold and a conduction pathway in the same place, and share no alignable sequence. The common ancestry is likely; the data that would reconstruct it is not in the sequences.

Structurally equivalent, phylogenetically distant. The iGluR pore is an inverted potassium-channel pore. This is a real homology at the level of the module, and it does not license putting an AMPA receptor and a Kir channel in one alignment.

Genuinely convergent, so far as anyone can tell. Connexins and the innexin/pannexin group build the same kind of large intercellular pore and are not detectably related; vertebrates run both systems and invertebrates only one [26,36]. “Not detectably related” is a statement about detection methods, and section 11 treats it as a hypothesis rather than a conclusion.

3. Selectivity

3.1 The potassium filter

The best-understood selectivity mechanism in biology is the potassium filter: a short, absolutely conserved sequence — T-x-G-Y-G — whose backbone carbonyls line a narrow pore and coordinate a dehydrated K⁺ ion in the same geometry water would [7,37,38]. Sodium, which is smaller, cannot be coordinated at that spacing and pays a dehydration penalty it does not recover; selectivity is achieved by a structure that fits the larger ion, which is the opposite of a sieve. The filter is recognisable in *Streptomyces* KcsA, in *Drosophila* Shaker and in human Kv1.1, and it is the one motif in this review that can be found in a raw sequence with no reference and no annotation [39].

3.2 The four-repeat locus

The sodium and calcium channels arrived at selectivity differently. Both are pseudo-tetramers — four homologous 6TM repeats in a single chain, the product of two rounds of internal duplication from a Kv-like ancestor — and their selectivity is set by one residue contributed by each repeat, at the position corresponding to the potassium filter. In sodium channels those four residues read D-E-K-A; in the high-voltage-activated calcium channels E-E-E-E; and swapping them converts one into the other [40].

This locus is the only reliable way to tell the four-repeat families apart. Their domain architectures are identical, their lengths overlap, and no domain model distinguishes them. NALCN, a sodium leak channel, reads E-E-K-E — a calcium-channel filter with a sodium-channel lysine in the third repeat [41]. The T-type calcium channels read E-E-D-D rather than E-E-E-E, which is a real difference within a family rather than an error.

Several of the families whose filters are least well characterised are also the primary transducers of noxious stimuli, which is where much of their pharmacological interest lies [42].

3.3 Selectivity that is not in the family

Two observations set a limit on how far family membership predicts what a channel conducts.

The first is that charge selectivity in the Cys-loop receptors — the difference between an excitatory nicotinic receptor and an inhibitory GABA or glycine receptor — is set by a short ring of residues at the intracellular end of the pore-lining helix, and can be inverted by three substitutions [43]. Cation- and anion-selective members sit in the same superfamily with the same fold.

The second is that AMPA-receptor calcium permeability is not encoded in the genome at all: it is set by RNA editing at a single position in GluA2, and a genomic census cannot see it [44].

The practical consequence is that selectivity is a property recorded per family from experiment, never predicted per sequence. Any pipeline that outputs a selectivity call from sequence alone is reporting family membership with extra steps.

3.4 Potassium selectivity twice

TMEM175, the endolysosomal potassium channel, is selective for K⁺ and has no T-x-G-Y-G filter and no structural relationship to the P-loop channels; its conduction pathway is lined by isoleucines [35,45]. The canonical filter is therefore one solution among at least two, and its near universality reflects the dominance of one superfamily rather than a physical necessity.

4. Gating

Gating is the property that most often gets mistaken for a clade. It is worth stating plainly: **no gating mechanism in this review is confined to one superfamily, and no superfamily is confined to one gating mechanism.**

Voltage. The canonical voltage sensor is a four-helix bundle whose fourth helix carries a repeating pattern of arginines and moves outward on depolarisation [46,47]. It is not exclusive to channels: the same domain is coupled to a phosphatase in the voltage-sensing phosphatases [48], and forms the entire conduction pathway of the voltage-gated proton channel, which has no pore domain at all [33,34,49]. Conversely, several P-loop families retain a complete sensor and are voltage-insensitive — the cyclic-nucleotide-gated channels among them [50] — and the HCN channels use the same apparatus inverted, opening on hyperpolarisation [51,52].

Extracellular ligand. Four unrelated superfamilies bind a transmitter outside and open: the pentameric Cys-loop receptors, the tetrameric glutamate receptors, the trimeric P2X receptors and, in a different sense, the proton-gated ASICs [18]. Their ligand-binding domains are unrelated clamshells, β -sandwiches and dolphin-shaped ectodomains respectively.

Intracellular ligand. Cyclic nucleotides bind a conserved cyclic-nucleotide-binding domain in CNG and HCN channels; the *same domain*, in a form that binds nothing, sits in the EAG/ERG potassium channels [50]. Calcium acts through an RCK ring in BK channels [53] and through constitutively bound calmodulin in the SK channels; IP₃ and ryanodine act on the huge cytosolic solenoids of the intracellular release channels [54,55].

Force. Mechanosensitivity has been invented at least six times: Piezo [56], the bacterial MscL and MscS — which are unrelated to each other [30,57] — the plant and animal OSCA/TMEM63 family [22,58], the TMC proteins of hair cells [59,60], the mechanosensitive two-pore-domain potassium channels [61], and the *Drosophila* NOMPC channel with its 29-ankyrin tether [62]. Some sense force from the lipid, some through a tether; the mechanism is a physical strategy, not a lineage [63].

Temperature, protons, lipids, volume. Thermal gating in the TRP families [64], proton gating in ASICs and Hv1, PIP₂ dependence across the inward rectifiers and KCNQ channels [65], and cell-volume gating in LRRC8 [66,67] complete the list without adding a clade.

4.1 The consequence for classification

None of this is recoverable from sequence family membership with any reliability. HCN and CNG channels have the same domain architecture and opposite voltage dependence. SK channels have an intact voltage sensor and are voltage-insensitive. A classifier can say what family a protein belongs to; it cannot say what opens it, and a review that blurs the two makes the literature harder to use rather than easier.

5. The P-loop superfamily

Roughly 143 of the ~320 human pore-forming genes sit in this one superfamily, and 79 of those are potassium channels [10,68]. It is the only division of the subject large enough that its internal structure is itself a research problem.

5.1 The potassium branch

Topology organises it [69]. The **6TM/1P** architecture — a voltage sensor plus a pore module — covers the Shaker-related Kv1–Kv4 channels [5,6], the KCNQ channels, the EAG/ERG/ELK family, the calcium-activated BK [53] and SK channels, and the electrically silent modifier subunits that cannot conduct alone. The **2TM/1P** inward rectifiers drop the sensor entirely and rectify through block by intracellular magnesium and polyamines [65,70,71]. The **4TM/2P** two-pore-domain channels are dimers carrying two pore loops per subunit, and supply the leak conductances that set resting potential [61,72].

Prokaryotic members root all of it: KcsA [7], the calcium-gated MthK [53], and KirBac [70] are not distant curiosities but the structural basis of the whole branch [73].

5.2 The four-repeat channels

Two rounds of internal duplication produced a single chain with four 6TM repeats: the voltage-gated sodium channels [4,74], the calcium channels [75,76], the sodium leak channel NALCN [41], and the sperm-specific CatSper complex, which is unusual in assembling its pseudo-tetramer from four *different* genes [77,78]. The two-pore channels of endolysosomes, with two repeats rather than one or four, are the intermediate the story predicts [79].

5.3 The TRP families

TRPC, TRPV, TRPM, TRPA, TRPML and TRPP share the pore fold and little else: their cytoplasmic domains range from a handful of ankyrin repeats to fourteen, and TRPM2, TRPM6 and TRPM7 are chanzymes — a channel fused to an enzyme [80,81]. The founding member came from a *Drosophila* phototransduction mutant [82], and the division as a whole is best understood as a set of cellular sensors rather than as a functional class [83]; TRPV1, cloned as the capsaicin receptor, became the first structure solved by single-particle cryo-EM at near-atomic resolution and effectively started that era [64,84]. TRPC2 is a pseudogene in humans and a functional vomeronasal channel in mice, which makes it a useful test of whether a census makes per-lineage presence calls correctly [85].

5.4 The intracellular calcium-release channels

The IP₃ and ryanodine receptors belong here structurally — their C-terminal pore is a P-loop module — and nowhere near here in size: RyR1 is 5,038 residues, the largest ion channel known, against ~400 for a Kir channel [54,55,86,87]. Almost all of the difference is cytosolic solenoid with no counterpart elsewhere in the superfamily, which is why any cross-family alignment has to be restricted to the pore module.

6. The ligand-gated superfamilies

Four unrelated superfamilies open when a transmitter binds. They are worth treating together only because their traps have the same shape: in each, the ligand-binding domain has an independent existence outside channels, and a search built on it returns proteins that do not conduct.

6.1 Pentameric (Cys-loop) receptors

Five subunits, each a β -sandwich ectodomain over four transmembrane helices, with the second helix lining the pore [11,88]. The human complement is large — nicotinic acetylcholine receptors, GABA_A receptors, glycine receptors, the ionotropic serotonin receptor and the zinc-activated channel — and the superfamily is alignable end to end, which makes it the most tractable phylogenetic unit in this review.

Two members are informative precisely because they are not what the family name implies. *AChBP*, secreted by snail glia, is a complete pentameric ligand-binding domain with no transmembrane region at all, and its structure made nicotinic pharmacology interpretable [12]. The bacterial channels GLIC and ELIC lack the disulfide-bonded loop the superfamily is named after [13,14] — the defining feature is not universal within the group it defines. Both facts matter for classification: a rule requiring the transmembrane domain alongside the ligand-binding domain correctly rejects AChBP and, as measured in section 12, also rejects the bacterial channels that root the tree.

Invertebrates add glutamate- and histamine-gated *chloride* channels absent from vertebrates — the ivermectin target — and one of them provided the first eukaryotic structure of the superfamily [89]. Human structures followed for GABA_A and glycine receptors [90,91].

6.2 Ionotropic glutamate receptors

Four subunits, each with an amino-terminal clamshell, a second clamshell that binds glutamate, and a pore module inserted the other way up [15,92,93]. The AMPA, kainate, NMDA and delta families differ in ligand requirements, kinetics and — for NMDA — the coincidence detection that comes from needing glutamate, glycine and depolarisation together [94,95].

The trap here is the clamshell. It is a bacterial periplasmic binding-protein fold, and it is also the ligand-binding domain of every class C G-protein-coupled receptor: the metabotropic glutamate receptors, both GABA_B subunits, the calcium-sensing receptor and the sweet and umami taste receptors. A search for

glutamate receptors by clamshell returns a receptor family that is not ionotropic at all, and the discriminating evidence is the pore region.

The superfamily also extends well past animals with synapses. Plants carry glutamate-receptor genes [96], and the insect ionotropic receptors are divergent iGluRs that serve chemosensation and have mostly lost the clamshell — so a clamshell-based search misses them entirely [97]. The delta receptors do not gate to glutamate at all, and are known to have a functional pore only because a gain-of-function mutation holds it open [98].

6.3 P2X receptors

Trimers gated by extracellular ATP, with two transmembrane helices per subunit, both termini intracellular, and a large disulfide-rich ectodomain shaped, in the original description, like a leaping dolphin [17,99]. Their distribution is patchy in an interesting way: present in amoebozoa — where a *Dictyostelium* member works on intracellular vesicles rather than the plasma membrane [100] — and absent from *Drosophila* and *C. elegans*, which makes P2X a good test of whether a census can distinguish gene loss from database absence.

6.4 DEG/ENaC channels

Trimers again, of an unrelated fold, and gated by almost everything except voltage: protons in the acid-sensing channels [18], peptides in the molluscan FaNaC, mechanical force in the *C. elegans* degenerins, and in the epithelial sodium channel essentially nothing — ENaC is constitutively open and regulated by proteolysis and trafficking [19,101]. The superfamily takes its name from gain-of-function alleles that kill the neuron they are expressed in, found in the first genetic screens for touch insensitivity [102]. Its size is strongly lineage-dependent: nine members in humans against roughly thirty in *C. elegans* and thirty pickpocket genes in *Drosophila*, so a human-only count badly misrepresents it.

7. Anion channels

Anion channels were the last major group to yield their molecular identities, and the delay shows in how many of them are still argued about. Chloride is the abundant permeant anion, its equilibrium potential sits close to rest in many cells, and small conductances are hard to attribute — so several activities were characterised physiologically for decades before the gene was found [103].

7.1 CLC: one family, two mechanisms

The CLC family is a homodimer in which each subunit contains a complete, independent pore — a “double-barrelled” architecture inferred from noise analysis long before the structure confirmed it [20]. What the structure did not predict is that half the family are not channels at all. The bacterial prototype ClC-ec1 is a $2\text{Cl}^-/\text{H}^+$ antiporter [8], and so are the mammalian endosomal and lysosomal members CLC-3 to CLC-7; only CLC-1, CLC-2 and the CLC-K channels conduct passively [104].

The mechanistic difference turns on a conserved glutamate on the intracellular side, and the transport phenotype is ancestral: channel behaviour is derived within a family whose prototype transports. For a census this is the cleanest possible demonstration that family membership and mechanism are separate questions with separate evidence, and that answering one does not answer the other.

7.2 CFTR: a transporter that is a channel

CFTR is an ABC transporter — two transmembrane domains, two nucleotide-binding domains, the whole architecture — that conducts anions down their gradient [31,105,106]. It is gated by ATP binding at the nucleotide-binding domain interface and licensed by PKA phosphorylation of a regulatory domain its siblings lack. Its nearest relatives, including the sulfonyleurea receptors that regulate the Kir6 potassium channels, are not channels.

CFTR is therefore the counter-example to every rule of the form “fold X implies transporter”. Any classifier that excludes proteins for carrying an ABC domain excludes the most clinically important chloride channel in the human genome; the discriminating evidence is one accession.

7.3 TMEM16: channels and scramblases on one fold

Three groups identified TMEM16A simultaneously as the long-sought calcium-activated chloride channel [107,108,109]. The structure of a fungal relative then showed something unexpected: the same fold, with the conduction pathway open to the lipid rather than closed into a pore, is a phospholipid scramblase [21]. Most of the family scrambles [9]; TMEM16A and TMEM16B conduct [110].

The two functions are not distinguishable by domain architecture, by length or by any sequence feature yet identified, which makes the channel / scramblase split a literature assignment rather than a measurement — and a hypothesis a phylogeny can test.

7.4 Bestrophins, tweety, and the volume-regulated channel

Bestrophins are pentamers with a single central pore, unrelated to anything else here, identified through a macular dystrophy gene [111,112].

The volume-regulated anion channel — a current known since the 1980s and central to cell-volume regulation — was assigned to two different protein families before two parallel genome-wide screens identified LRRC8 in 2014 [27,66,67]. The tweety proteins, one of the earlier candidates, remain in catalogues as contested: recent structures show no obvious conduction pathway. Keeping them, flagged, is the honest option; dropping a family because the literature moved makes a census a record of opinion.

8. Large pores

A pore wide enough to pass ATP is a different object from a filter that discriminates K^+ from Na^+ by a fraction of an ångström, and the two are easily conflated by a definition that only asks whether ions cross. The large-pore channels are included here because they are gated aqueous pathways, and flagged because no statement about ionic selectivity should include them.

Connexins form hexameric hemichannels that dock across an extracellular gap into a complete intercellular channel [25,113]. There are 21 human genes, named twice over — by Greek-letter subfamily (GJA1) and by predicted molecular weight (Cx43) — with the two systems not in register, which is a pure nomenclature hazard. GJB2 mutations are the commonest cause of inherited deafness.

Innexins do the same job in invertebrates and are unrelated to connexins [36]. **Pannexins** are the vertebrate members of the innexin family; they are glycosylated, so they do not dock into junctions, and work instead as regulated ATP-release conduits [26].

LRRC8 completes the group: its transmembrane region is homologous to pannexin — Pfam names the domain “Pannexin-like TM region of LRRC8” — with a leucine-rich-repeat domain added [27]. Substrate selectivity, including cisplatin uptake, is set by which of LRRC8B–E join the obligatory LRRC8A.

CALHM is a fourth, unrelated solution: a large pore of variable oligomeric state that releases ATP from taste cells, and is the transmitter release mechanism for sweet, bitter and umami taste [28,114].

8.1 A convergence with a twist

Vertebrates run two independent intercellular-channel systems. One — innexin/pannexin, with LRRC8 attached — they share with invertebrates. The other, the connexins, they appear to have invented; no connexin has been found outside chordates, and no sequence relationship to the innexin group is detectable [26,36].

This is the best-supported case of fold-level convergence in the subject. It is also, precisely because it rests on a negative, the one most worth re-testing with profile-profile and structural methods rather than sequence search: “not detectably related” is a statement about the method used to look.

9. Mechanosensation

Turning force into a conductance is the clearest case in this review of one problem solved repeatedly by unrelated proteins. At least six architectures do it, and they share nothing but the physics [63].

Piezo1 and **Piezo2** are three-bladed propellers of 38 transmembrane helices per subunit — the largest transmembrane count of any known channel — that bow the surrounding membrane into a dome and

flatten under tension [23,56]. PIEZO2 carries touch and proprioception; PIEZO1 loss-of-function causes lymphatic dysplasia and gain-of-function a haemolytic anaemia.

MscL and MscS are the bacterial safety valves, and they are unrelated to each other as well as to everything else. MscL is a 136-residue pentamer with the largest conductance known, opening under near-lytic tension to release osmolytes [29,57]; MscS is a heptamer with a lipid-facing gate and a large cytoplasmic vestibule [30]. The family is present in bacteria, archaea, plants, fungi and protists and absent from animals — one of the sharpest presence/absence claims in the subject, and one that deserves testing rather than repeating.

OSCA/TMEM63 were found in plants as hyperosmolality-gated calcium channels [58] and only then recognised in animals as an evolutionarily conserved family of mechanically activated channels [22]. They share the TMEM16 fold without sharing its sequence — a relationship visible only in structure. Their history is a caution: a family that had been present in animal genomes all along was found in plants first, and a human-only survey would have called it absent.

TMC1 and TMC2 are the strongest candidates for the pore of the hair-cell mechanotransduction channel, the fastest sensory transduction in vertebrates [59,60]. The evidence is genetic, physiological and structural-by-homology; there is still no structure of a conducting state, which is why careful catalogues list them as contested. TMC6 and TMC8 do something else entirely in keratinocytes.

The mechanosensitive potassium channels — TRAAK and TREK, in the two-pore-domain family — are gated by membrane tension through a lipid-facing opening in an otherwise ordinary P-loop architecture [61]. Force gating here is a property of one branch of a superfamily whose other members are gated by everything else.

NOMPC, a *Drosophila* TRP channel with 29 ankyrin repeats forming a spring between the channel and the cytoskeleton, is the clearest tethered mechanotransducer known [62]. It was lost in mammals.

9.1 Why this matters for a census

Mechanosensitivity is a mechanism, not a lineage. A catalogue organised by what opens a channel would place Piezo, MscS, OSCA, TMC, TRAAK and NOMPC in one group and would be wrong about every evolutionary question it was then asked. A catalogue organised by fold keeps them apart and records the shared physiology as an annotation — which is the only arrangement in which the question “how many times has this been invented?” can even be posed.

10. Channels of intracellular membranes

Organelle channels are systematically under-represented in channel censuses, for a reason worth naming: they were found by physiology on isolated membranes rather than by patch clamp on cells, so they entered the literature later, with less standardised nomenclature, and several still have contested molecular identities. A census whose assumptions are shaped by plasma-membrane channels — a signal peptide, an extracellular ligand-binding domain, a voltage sensor — finds fewer of them than exist.

IP₃ receptors and ryanodine receptors release calcium from the endoplasmic and sarcoplasmic reticulum. Both are homotetramers of enormous subunits — 2,700 residues for an IP₃ receptor, 5,038 for RyR1, the largest ion channel known — in which a P-loop pore module sits atop a cytosolic solenoid that acts as a signal-integrating scaffold [54,55,86,87,115]. They share four Pfam domains and are the textbook case of two families that domain annotation cannot separate.

TRIC channels (TMEM38A/B) are trimeric monovalent cation channels of the SR/ER membrane whose entire function is to provide the counter-ion flux that lets the release channels work without building an opposing voltage [116]. A channel whose job is to make another channel possible is easy to leave out of a list of channels.

The mitochondrial calcium uniporter carried a measured current for fifty years before its gene was found, by two groups simultaneously using comparative genomics [117,118]. It is the most calcium-selective channel known. Its regulators — MICU1–3, EMRE, MCUB — are a good test of the pore/accessory distinction: MCUB is a pore subunit, EMRE is required for conduction and lines no pore.

VDAC is the main conduit across the outer mitochondrial membrane and the only all- β channel in the subject: a 19-stranded barrel, an odd number that forces one parallel β -pair, unlike any bacterial porin [24]. It is gated by voltage and regulated, which is what keeps it inside the definition while ungated bacterial porins fall outside.

TMEM175 is the endolysosomal potassium channel, and it is selective for K^+ without a T-x-G-Y-G filter — its pathway is lined by isoleucines [35,45]. It is also a Parkinson's disease risk locus. Together with the canonical filter it establishes that potassium selectivity has evolved at least twice, which is the single most useful fact in this review for calibrating how much weight a motif can carry.

ORAI channels are not intracellular, but belong beside them: the store-operated calcium current is gated by physical contact with STIM proteins in the ER membrane across a junction, so the gate for a plasma-membrane channel sits in another organelle [119,120,121,122].

Two-pore channels (TPC1/2) sit in endolysosomal membranes, are gated by $PI(3,5)P_2$ and NAADP, and have two 6TM repeats rather than one or four — exactly the intermediate the duplication history of the four-repeat channels predicts [79].

11. Evolution

11.1 Channels are older than the things they are famous for

Voltage-gated channels are usually introduced through the action potential, which invites the assumption that they arose with nervous systems. They did not. Potassium channels are present across bacteria and archaea, and the prokaryotic structures are not distant analogues but the direct structural basis of the eukaryotic families [73,123]. Sodium-channel homologues predate the origin of nervous systems in animals [124], and the voltage-gated set was already diversifying at the emergence of the Metazoa [125]. Sponges and placozoans, which have no neurons, carry substantial channel repertoires; the anemone *Nematostella* carries a nearly complete one.

The ordering is therefore the reverse of the intuitive one. The channels came first and the nervous system was assembled from parts that already existed, which is why a census restricted to animals with synapses answers the wrong question.

11.2 Repertoires are lineage-specific in both directions

Comparative work on complete genomes shows expansions and losses that correlate with lifestyle rather than with taxonomic rank [68,126,127]. *C. elegans* has roughly ninety potassium channel genes; humans have about seventy-nine. The first genome-scale survey of a channel repertoire, in *Drosophila*, already made the point that the set is not a scaled-down version of the vertebrate one [128]. The DEG/ENaC superfamily has nine human members against roughly thirty in *C. elegans* and thirty pickpocket genes in *Drosophila*. P2X receptors are present in amoebozoans and absent from both major invertebrate models [100]. TRPC2 is a functional vomeronasal channel in mice and a pseudogene in humans [85]. NOMPC, the clearest tethered mechanotransducer known, was lost in mammals [62].

Plants make the point from the other side. They have no voltage-gated sodium channels and no nervous system, and they have glutamate receptors [96], two-pore channels, mechanosensitive MscS-like channels and the OSCA family that was found in plants before it was recognised in animals [22,58,129].

11.3 Convergence, and the limits of the claim

Four convergences are well enough supported to state, with the caveat that each rests on failure to detect homology:

- **Potassium selectivity**, achieved by the canonical filter and, independently, by TMEM175's isoleucine-lined pathway [35].
- **Intercellular channels**, built by connexins in chordates and by the unrelated innexin/pannexin family elsewhere [26,36].
- **Mechanosensitivity**, invented at least six times (section 9).
- **Large-pore ATP release**, by pannexin, LRRC8 and CALHM on at least two unrelated folds [27,114].

The honest formulation is that these are cases where sequence methods find no relationship. Structure comparison and profile-profile search can push the detection limit further, and the TMEM16/OSCA/TMC clan — three families assigned separate domain models that turned out to share a fold — is the warning that they sometimes do.

11.4 What a phylogeny of ion channels can and cannot be

There is no alignment containing a nicotinic receptor and a Kv channel, so there is no tree of ion channels. What is well defined is a **forest**: within-family trees, which are conventional and where most biologically useful results live; within-superfamily trees, which for the P-loop superfamily must be built on the pore module alone, because members range from 400 to 5,038 residues and almost all of the difference is cytosolic machinery with no counterpart across families.

Between superfamilies the correct object is a **network** of structural similarity, with no branch lengths and no support values, in which an edge records measured fold similarity and asserts no common ancestor. Every published tree spanning non-homologous channel superfamilies is measuring alignment artefacts.

12. What the databases record

Everything above is drawn from the primary literature. This section is different: it reports what the annotation databases actually contain about these proteins, measured directly against InterPro and UniProt on 2026-08-19 with the queries recorded in `results/s0_baseline/` [130,131,132]. It is included because every computational statement about ion channels — every census, every “how many are there”, every claim of absence in a lineage — passes through these resources, and their coverage is not uniform in ways that are rarely stated.

The measurements below come from a catalogue of 90 families in 25 superfamilies, of which 68 are ion-channel families covering 320 human pore-forming genes, with 115 declared domain accessions and 162 reference proteins. All 115 accessions and all 162 exemplars resolved.

12.1 A pore model is not a family, and not always a pore

PF00520 (Ion_trans) is the canonical model of the P-loop pore module. In this catalogue it is carried by **20 different families**, which is expected — it is a superfamily-level model. Two of those families are the informative ones:

- **HVCN1**, the voltage-gated proton channel, carries it and *has no pore domain at all*: its conduction pathway is the voltage sensor [33,34].
- **TPTE**, a voltage-sensing phosphatase, carries it and is not a channel [48].

So the presence of the canonical pore model is not evidence that a protein has a pore, let alone which family it belongs to.

12.2 Most TRP families carry no copy of it

Measured across exemplars: TRPV1, TRPV5, TRPA1 and TRPM2 carry PF00520; **TRPC3, TRPM8, MCOLN1, MCOLN2 and PKD2 carry none of it**. Their pore regions are covered by different models entirely — PF08344 for TRPC, PF18139/PF25508 for TRPM, PF08016 for the mucolipins and polycystins.

A census that enumerates P-loop channels from PF00520 therefore loses most of the TRP division without any error message. This is the single most consequential annotation fact we measured, and it is not a database error: the models were built family by family from the best-characterised members, and behave accordingly.

12.3 Domain models do not track topology

The small-conductance calcium-activated potassium channels are 6TM proteins, and InterPro annotates KCNN2's pore with PF07885 — the 2TM model. So does KCNT1, while its close relative KCNMA1 gets PF00520. Any rule that infers topology from which pore model matched is wrong for these families; topology has to come from transmembrane predictions or experimental annotation.

Similarly, the inward rectifiers carry **no** PF07885 at all — they are covered by PF01007 and two cytoplasmic-domain models. The textbook grouping of Kir with the two-pore-domain channels as “2TM/1P potassium channels” has no counterpart in the annotation.

12.4 Architecturally identical families

Four cases where domain composition cannot distinguish families that are functionally distinct:

families	shared architecture	what separates them
Nav, Cav, NALCN, CatSper	PF00520×4	the four filter residues (§3.2)
CFTR vs ABCC8/9	PF00664×2 + PF00005×2	PF14396, CFTR's R domain
TMEM16A/B vs the scramblases	PF04547 + PF16178	nothing known
TRPML vs TRPP vs polycystin-1	PF08016	PF21381 vs PF20519 vs PKD repeats

The third row is the uncomfortable one: ANO1 and ANO6 have identical architectures, ~40 % identity and different functions, and no sequence-level test yet separates them.

12.5 Signatures shared with non-channels

Of 26 domain accessions carried by more than one family in this catalogue, **15 cross the channel / non-channel boundary**. Beyond PF00520:

- PF02214, the Kv T1 tetramerisation domain, is a generic BTB/POZ domain carried by 25 human KCTD proteins that are mostly ubiquitin-ligase adaptors.
- PF01094, the iGluR clamshell, is the ligand-binding domain of every class C GPCR (§6.2).
- PF02931, the Cys-loop ligand-binding domain, is a complete soluble protein in AChBP [12].
- PF02815, the MIR domain of the intracellular release channels, is carried by the protein O-mannosyltransferases.

12.6 The outgroups are the worst-covered proteins in the subject

The finding with the most consequence for phylogenetics: **three of the four prokaryotic outgroups that root the major superfamily trees cannot be found by the domain rules that define their own superfamilies**.

- **GLIC and ELIC**, which root the Cys-loop tree, carry PF02931 and no PF02932 — so a rule requiring the transmembrane domain alongside the ligand-binding domain, which correctly rejects AChBP, also rejects them [13,14].
- **GluR0**, the cyanobacterial glutamate receptor that roots the iGluR tree, is annotated as PF07885 + PF00497 — *a potassium-channel pore fused to a bacterial solute-binding protein* — with none of the eukaryotic iGluR models [16]. That is the inverted-P-loop story visible directly in the annotation, and it means no iGluR-signature search will ever find it.
- **NavAb**, the prototype prokaryotic sodium channel, carries neither PF06512 nor PF11933, the two Nav-specific models: it is the pore and nothing else [74].

Seven of 162 reference proteins have **no gene name in UniProt at all**, and they are almost all prokaryotic or invertebrate: NavAb, ELIC, GluR0, FaNaC, the *Bacillus* NaK channel, the *Klebsiella* bestrophin and AChBP. One, GLIC, was invisible to a symbol-based search because its entry is filed under a bacterial strain rather than a species.

12.7 What follows

None of this argues against using these resources; there is no alternative, and their coverage of well-studied families is excellent. It argues for three specific practices. Enumerate from the union of pore models rather than the canonical one. Treat a domain hit as evidence at the level where the model is diagnostic and no further. And expect the proteins that root your trees to be the ones your search cannot find.

13. Disease and pharmacology

13.1 Channelopathies

The ion channels carry the largest set of Mendelian diseases attached to any protein family, and the pattern is instructive: because a channel's output is an electrical or chemical flux, small changes in gating kinetics

produce large, tissue-specific phenotypes, and both loss and gain of function are pathogenic in ways that are often opposite [133,134].

The list spans nearly every family in this review. CFTR in cystic fibrosis [31]; SCN1A in Dravet syndrome and SCN5A in Brugada and long-QT syndromes; KCNQ1 and KCNH2 in the commonest inherited arrhythmias [135]; RYR1 in malignant hyperthermia and central core disease; CLCN1 in myotonia; GJB2 as the commonest cause of inherited deafness; PKD1 and PKD2 in autosomal dominant polycystic kidney disease; KCNJ11 in neonatal diabetes; SCNN1A in Liddle syndrome; BEST1 in Best vitelliform macular dystrophy [111]; TMC1 in inherited deafness [59]; DRAI1 in a severe combined immunodeficiency [119]; PIEZO1 in lymphatic dysplasia and hereditary xerocytosis [56]; MCOLN1 in mucopolipidosis IV; TMEM38B in a recessive osteogenesis imperfecta [116]; and TMEM175 as a Parkinson's disease risk locus [35].

Two features of this list matter for a computational census. First, its breadth is a reason to get family assignment right: a variant's interpretation depends on which channel it is in. Second, the diseases cluster in the *best-studied* families, which is a statement about ascertainment as much as about biology — and any analysis correlating disease burden with family properties has to say so.

13.2 Drug targets

Ion channels are among the most heavily drugged protein classes, second only to GPCRs by most counts [136,137]. Local anaesthetics, class I antiarrhythmics and many anticonvulsants act on voltage-gated sodium channels; dihydropyridines and related agents on L-type calcium channels; sulfonylureas on the Kir6/SUR complex; benzodiazepines, barbiturates and most general anaesthetics on GABA_A receptors; ivermectin on invertebrate glutamate-gated chloride channels; amantadine on the influenza M2 viroporin; gabapentin and pregabalin on the calcium-channel $\alpha 2\delta$ subunit — which is an auxiliary subunit and not a pore [138,139].

The last example is worth dwelling on. One of the most prescribed “ion-channel drugs” targets a protein that this review's definition excludes from the channel census. That is not an argument for changing the definition; it is an argument for keeping the accessory subunits in the catalogue, marked, so that the question can be asked at all.

Systematic target annotation is maintained by IUPHAR/BPS, whose concise guide is the reference nomenclature for the field [140]. Where its family boundaries differ from the ones used here, the difference is almost always one of the six scope decisions in section 1.

14. Open questions

The questions below are the ones this review could not answer from the literature, phrased so that a computational census could.

Q1 — How many ion-channel genes are there, and how much of the published range is scope? Human counts run 240–400. Almost all of the spread is the six decisions in section 1, not disagreement about biology, but nobody has recomputed the total under each. A census that reports its answer once per scope decision would replace an argument with a table.

Q2 — Can family assignment be made without reading gene names, and where does it fail? Nomenclature encodes the order of discovery, and unnamed gene models in new genomes have none. A classifier built only on positive sequence and structural evidence would show which families are recoverable and which depend on the annotation already being right.

Q3 — Which superfamilies are reachable by domain search at all? Section 12 shows that most TRP families and three of the four superfamily outgroups are not. The recall of domain search, profile search and genomic search, measured per superfamily, would set an upper bound on every existing census.

Q4 — Is the absence of MscS-family channels from animals a genome fact or a database fact? It is repeated confidently and rests on annotation coverage in exactly the lineages where coverage is weakest.

Q5 — Do selectivity filters track phylogeny? The T-type calcium channels read E-E-D-D where their relatives read E-E-E-E. Mapping the filter onto a pore-module tree would show whether selectivity follows descent or has been re-tuned repeatedly.

Q6 — Is connexin/pannexin convergence real, or homology below the detection limit? The claim rests on a negative from sequence search. Profile-profile comparison and structure alignment can move that limit, and the TMEM16/OSCA/TMC clan shows they sometimes do.

Q7 — Do the mechanism-based splits survive a phylogeny? CLC channels against CLC transporters, TMEM16 channels against TMEM16 scramblases. If each mechanism forms a clade, the split is evolutionary; if it cuts across clades, the split is physiology imposed on a family that does not respect it.

Q8 — How much apparent lineage-specific loss is assembly and gene-set quality? Every claim in section 11.2 rests on absence in a particular gene set, and gene sets differ in provenance and completeness far more than the claims usually acknowledge.

15. How the claims in this review were checked

This review is generated, and its verification is mechanical rather than editorial. Three checks run on every build.

Every reference was resolved against a live literature service. The source list (`scripts/review_sources.py`) contains *titles*, not citations. `scripts/s0_review_refs.py` queries Europe PMC for each, accepts a record only when its title is at least 90 % similar to the one requested, and writes the returned PMID, DOI, year, journal and authors to `results/s0_baseline/references.tsv`. A title that does not resolve is reported and not written.

That check has teeth. On the first run, seven titles failed: five were short generic titles where the search ranked another paper, and two were misremembered — the Cav1.1 structure paper is *Structure of the voltage-gated calcium channel Cav1.1 complex* (Science, 2015), and the Moran review is *Evolution of voltage-gated ion channels at the emergence of Metazoa*, not “of Nervous Systems”. Both were corrected against the returned records. The final list is 148 of 148 resolved.

A citation with no verified reference is a build failure. `scripts/s0_review_build.py` collects every [key] in the section files, rennumbers them into order of first appearance, and refuses to write the document if any key is missing from the reference table. Prose therefore cannot cite a paper that was never resolved.

Every [db] statement is re-derivable. Section 12 reports measurements rather than literature, and the queries that produced them are in `scripts/s0_catalogue_verify.py`; the tables are in `results/s0_baseline/`. Re-running the script re-derives every number in that section, and the S0 report is rendered from the tables rather than written alongside them.

The computational toolchain

Where this review reports its own measurements, the tools are: MAFFT for alignment [141], trimAl for alignment trimming [142], IQ-TREE 2 with ModelFinder and ultrafast bootstrap for phylogenies [143,144], HMMER for profile searches [145], Foldseek for structure comparison [146], and the AlphaFold database for predicted structures where experimental ones are absent [147,148]. Domain and sequence annotation comes from InterPro, Pfam and UniProt [130,131,132].

What this does not verify

Resolution confirms that a paper exists with the title cited. It does not confirm that the paper says what the sentence citing it claims. That remains an editorial responsibility, and the mechanical checks should not be mistaken for having discharged it.

Nor does it verify the *selection*: which papers are cited, and which landmark results are missing, is a judgement, and a review assembled by searching for known titles will under-represent work the assembler did not already know about. The bibliography here is weighted towards structures, identifications and reviews in high-visibility journals, and that weighting is a property of how it was built.

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